

Domain & SSL Certificate Expiry Alert System – Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall initiate TLS handshakes against monitored domains daily, extract certificate expiration dates, and trigger alerts at 30, 15, and 3 days before expiry.	high	Pass: Alert queued for a certificate expiring in 15 days. Fail: Expired certificate ignored without notification.	Prevents SSL/TLS certificates from expiring unexpectedly.
FR-002	Functional	The system shall perform daily WHOIS registration audits and retrieve the expiry date of each monitored domain.	High	Pass: The system successfully records the registration expiry date for a monitored domain. Fail: No expiry information is recorded.	Prevents domain registration from expiring unexpectedly.
FR-003	Functional	The system shall allow the SysAdmin to add, remove, and update monitored domains and SSL/TLS endpoints.	High	Pass: A valid added or updated endpoint is saved and appears in the monitoring list. Fail: The valid change is not saved.	Allows the SysAdmin to manage the domains and endpoints being monitored.
FR-004	Functional	The system shall notify the SysAdmin and Security Officer when a domain or SSL certificate reaches a	High	Pass: A notification is generated when an item reaches the 30, 15, or 3-day threshold.	Ensures stakeholders are informed before expiry.

		configured expiry alert threshold.		Fail: No notification is generated.	
FR-005	Functional	The system shall escalate unresolved expiry alerts to the Security Officer when the expiry reaches a higher-priority threshold.	Medium	Pass: An unresolved alert is escalated when the configured escalation condition is met. Fail: The alert is not escalated.	Ensures critical expiry risks receive additional attention.
FR-006	Functional	The monitoring engine shall scan a list of 1,000 domains and SSL endpoints in under 3 minutes.	High	Pass: Benchmarking confirms completion in under 3 minutes under simulated peak load. Fail: The scan takes 3 minutes or more.	Ensures efficient monitoring of a large number of endpoints.
FR-007	Functional	The system shall record audit failures and continue processing the remaining monitored domains if an individual WHOIS query or TLS handshake fails.	High	Pass: When one endpoint fails, the failure is logged and the remaining endpoints continue to be processed. Fail: One failed	Ensures reliable and continuous monitoring

				endpoint stops the entire audit.	
--	--	--	--	---	--